

Project	AssetTrack — IT Asset Management System
Frontend Stack	Next.js · Tailwind CSS · TanStack Query v5 · Zod · React Hook Form
Base API	http://localhost:8080/api/v1
Auth	Bearer JWT — Authorization header
Roles	DEVELOPER · MANAGER · ADMIN
State Pattern	URL-as-State (filters) + AuthContext (identity) + TanStack Query (server)

Table of Contents

1. Global Architecture & Conventions
2. Authentication (auth)
3. Dashboard (dashboard)
4. User Management (users)
5. Asset Management (assets)
6. Condition Reports (condition-reports)
7. Reports & Analytics (reports)
8. Notifications (notifications)
9. Role Permission Matrix
10. API Error Code Reference

1. Global Architecture & Conventions

1.1 Tech Stack

Next.js App Router	Routing — /auth/*, /dashboard, /users/*, /assets/*, /condition-reports/*, /reports
Tailwind CSS	Styling — utility-first, no global CSS sheets
React Hook Form + Zod	Form state & validation — zodResolver bridges schemas to forms
TanStack Query v5	Server state — useQuery (reads), useMutation (writes), cache invalidation
URL Search Params	UI filter & pagination state — URL is single source of truth
React Context API	Global identity — AuthContext stores token + user profile (id, name, role)
Lucide React	Icon library

1.2 Standard API Envelope

Every API response is wrapped in a consistent envelope. Clients must check **success** first, then read **data** (success) or **error** (failure). Paginated responses include **data.items[]** and **data.pagination** { page (0-indexed), size, totalElements, totalPages }.

1.3 Zero-Local-State Error Pattern

Rule: Never use useState to track server errors in forms. All mutation errors are mapped through React Hook Form's setError:

- Field errors (409/422) → setError("fieldName", { message })
- Auth errors (401) → setError("root", { message }) — never field-specific (prevents enumeration)
- Server errors (500) → setError("root.serverError", { message })

1.4 URL-as-State (Filters & Pagination)

Rule: Filter and pagination state live exclusively in URL query params via useSearchParams(). Zod schemas coerce URL strings to typed values and translate UI 1-indexed pages to API 0-indexed pages. Updates use router.replace() — triggering automatic TanStack Query refetch via the query key.

2. Authentication (auth)

Handles user identity, secure token storage, and initial application entry. Routes: /auth/login and /auth/register.

2.1 API Endpoints

Method	Endpoint	Roles	Description
POST	/auth/register	Public	Register a new user. Returns user data (no token). Redirect to login on success.
POST	/auth/login	Public	Authenticate and receive JWT. Returns accessToken + user { id, name, role }.

2.2 Functional Requirements

ID	Requirement	Roles	Priority
FR-A01	The /auth layout must be strictly presentational — centered card, branding only, no route protection logic.	All	MUST
FR-A02	Registration form must collect: name (1–120 chars), email (valid format, lowercased before submit), password (min 8 chars, 1 upper + 1 lower + 1 number + 1 special).	All	MUST
FR-A03	API must not be called if client-side Zod validation fails (email format, password complexity).	All	MUST
FR-A04	On successful registration (201), redirect to /auth/login. Do NOT attempt to extract a token — the register response contains no accessToken.	All	MUST
FR-A05	On 409 EMAIL_CONFLICT during registration, map error to setError("email", { message: "An account with this email already exists." }).	All	MUST
FR-A06	Login form must collect email (valid format) and password (non-empty).	All	MUST
FR-A07	On successful login (200), extract accessToken and user { id, name, role } from data. Call AuthContext.login(token, user). Store JWT and user profile in localStorage.	All	MUST
FR-A08	After successful login, navigate to router.push("/dashboard").	All	MUST
FR-A09	On 401 UNAUTHORIZED during login, map strictly to setError("root", { message: "Invalid email or password" }). Never map to a specific field.	All	MUST
FR-A10	On 500 INTERNAL_ERROR, map to setError("root.serverError", { message: "..." }).	All	MUST

3. Dashboard (dashboard)

The secure primary navigation hub. Adapts UI based on user role (RBAC).

3.1 API Endpoints

Method	Endpoint	Roles	Description
GET	/reports/dashboard	ADMIN, MANAGER	Live KPI data for AdminManagerView.
GET	/notifications	All auth	Fetch unread notifications for NotificationBell.

3.2 Functional Requirements

ID	Requirement	Roles	Priority
FR-D01	Dashboard layout.tsx must verify auth via AuthContext on every render. If no valid session, redirect to /auth/login immediately.	All	MUST
FR-D02	The global Navbar must be rendered by dashboard/layout.tsx and include: company branding, dynamic user greeting, NotificationBell, and user profile dropdown with logout.	All	MUST
FR-D03	Logout action must clear AuthContext state and localStorage, then redirect to /auth/login.	All	MUST
FR-D04	page.tsx must read user.role from AuthContext and render DeveloperView for DEVELOPER role, AdminManagerView for ADMIN or MANAGER role.	All	MUST
FR-D05	AdminManagerView must display live KPI cards from GET /reports/dashboard: total assets, open condition reports count, warranty expirations in 30 days.	ADMIN, MANAGER	MUST
FR-D06	AdminManagerView must display navigation cards to: /users (Personnel), /assets (Inventory), /condition-reports (Reports), /reports (Analytics).	ADMIN, MANAGER	MUST
FR-D07	"Assign Roles" navigation or UI affordance in AdminManagerView must be hidden/disabled unless user.role === "ADMIN".	MANAGER	MUST
FR-D08	DeveloperView must show navigation to: My Equipment (/assets?assignedUserId={id}), Spare Laptops, Report Issue, My Profile.	DEVELOPER	MUST
FR-D09	NotificationBell must display an unread count badge. Clicking opens a dropdown of latest notifications. Polling interval: 60 seconds via refetchInterval.	All	MUST

4. User Management (users)

Personnel directory and individual profiles. Strict RBAC via layout guards and UI element rendering.

4.1 API Endpoints

Method	Endpoint	Roles	Description
GET	/users	ADMIN, MANAGER	List all users. Params: page, size, role.
GET	/users/{id}	ADMIN, MANAGER, self	Get a single user profile including assignedAssets.
PATCH	/users/{id}/role	ADMIN only	Update user role. Body: { role: ADMIN MANAGER DEVELOPER }.
DELETE	/users/{id}	ADMIN only	Delete a user account.

4.2 Functional Requirements

ID	Requirement	Roles	Priority
FR-U01	users/layout.tsx must redirect DEVELOPER attempting to access /users (list) to /dashboard.	DEVELOPER	MUST
FR-U02	DEVELOPER accessing /users/[id] must be granted access only if id === currentUser.id (own profile). Otherwise redirect to /dashboard.	DEVELOPER	MUST
FR-U03	UserTable.tsx must completely hide "Edit Role" and "Delete" action buttons unless user.role === "ADMIN". Managers can view but not modify.	MANAGER	MUST
FR-U04	All filter/pagination state must live in the URL (useSearchParams). No useState for filters. Zod schema translates UI page=1 to API page=0.	ADMIN, MANAGER	MUST
FR-U05	Filter bar must support: role (ADMIN MANAGER DEVELOPER), page, size. Zod validates page >=0 and size 1–100.	ADMIN, MANAGER	MUST
FR-U06	TanStack Query filter object (from URL) must be the query key — URL changes trigger automatic background refetch.	ADMIN, MANAGER	MUST
FR-U07	Role update (PATCH /users/{id}/role) must use UpdateRoleModal. Role schema must be enum: ADMIN MANAGER DEVELOPER.	ADMIN	MUST
FR-U08	On 409 LAST_ADMIN_PROTECTION, map to setError("root", { message: "Cannot remove the last admin." }).	ADMIN	MUST
FR-U09	Delete user (DELETE /users/{id}): on 409 USER_HAS_ACTIVE_ALLOCATIONS show specific error "Return the user's assigned assets first". On 422 (self-delete), block the action.	ADMIN	MUST
FR-U10	On any successful mutation (role update, delete), call queryClient.invalidateQueries for ["users"] and ["user", id] to instantly refresh the UI.	ADMIN	MUST

5. Asset Management (assets)

Manages lifecycle of hardware (Laptops, Monitors, Accessories) and their allocation.

5.1 API Endpoints

Method	Endpoint	Roles	Description
GET	/assets	ADMIN, MANAGER	List assets with filters: type, status, brand, assignedUserId, warrantyExpiresBefore, page, size.
POST	/assets	ADMIN, MANAGER	Register a new asset.
GET	/assets/{id}	ADMIN, MANAGER, DEVELOPER (own)	Full asset details including allocationHistory and conditionReports.
PUT	/assets/{id}	ADMIN, MANAGER	Update asset metadata.
DELETE	/assets/{id}	ADMIN only	Decommission/delete an asset.
GET	/assets/spare-laptops	All auth	Available spare laptops.
GET	/assets/search	ADMIN, MANAGER	Free-text search across serial, brand, model.
GET	/assets/{id}/allocations	ADMIN, MANAGER	Full allocation history for an asset.
POST	/assets/{id}/allocations	ADMIN, MANAGER	Assign asset to user. Body: { userId }.
DELETE	/assets/{id}/allocations	ADMIN, MANAGER	Return (unassign) asset.

5.2 Functional Requirements

ID	Requirement	Roles	Priority
FR-AS01	assets/layout.tsx must verify auth. For DEVELOPER accessing /assets/[id], verify via API/cache that asset.assignedTo.id === currentUser.id — otherwise redirect to /dashboard.	All	MUST
FR-AS02	page.tsx renders AdminAssetView for ADMIN/MANAGER (full inventory + all actions) and DeveloperAssetView for DEVELOPER (own assets + spare laptops, condition reports only).	All	MUST
FR-AS03	All asset list filtering and pagination must live in URL. Filters: type, status, brand, assignedUserId, warrantyExpiresBefore, page, size. Zod coerces and validates all params.	ADMIN, MANAGER	MUST
FR-AS04	AssetFormModal (Create/Update) must block submission via Zod until: brand ≤80 chars, model ≤120 chars, serialNumber ≤100 chars, purchaseDate ≤ today, warrantyExpirationDate ≥ purchaseDate.	ADMIN, MANAGER	MUST
FR-AS05	On 409 SERIAL_NUMBER_CONFLICT, map to setError("serialNumber", { message: "Serial number already exists." }).	ADMIN, MANAGER	MUST

ID	Requirement	Roles	Priority
FR-AS06	AllocationModal must handle assign (POST) and return (DELETE) actions. On 409 ASSET_ALREADY_ASSIGNED or NO_ACTIVE_ALLOCATION, map to setError("root", ...).	ADMIN, MANAGER	MUST
FR-AS07	Asset Detail page (/assets/[id]) must display allocationHistory table from GET /assets/{id}/allocations (ADMIN/MANAGER only) and conditionReports list.	ADMIN, MANAGER	MUST
FR-AS08	DeleteAssetCard must be rendered only if user.role === "ADMIN". A confirmation dialog is required before firing the delete mutation.	ADMIN	MUST
FR-AS09	Asset search (/assets/search) must support free-text q param + type, status, assignedUserId, brand, warrantyExpired (boolean) filters. Separate from column filters.	ADMIN, MANAGER	MUST
FR-AS10	GET /assets/spare-laptops is accessible by all authenticated users including DEVELOPER (from DeveloperView).	All	MUST

6. Condition Reports (condition-reports)

Allows Developers to report hardware issues on assigned assets. Admins and Managers review, update status, and resolve reports.

6.1 API Endpoints

Method	Endpoint	Roles	Description
POST	/assets/{id}/condition-reports	DEVELOPER (own), ADMIN, MANAGER	Submit a new condition report.
GET	/condition-reports	ADMIN, MANAGER	List all reports. Params: status, severity, assetId, page, size.
PATCH	/condition-reports/{id}	ADMIN, MANAGER	Update status and/or resolution.

6.2 Functional Requirements

ID	Requirement	Roles	Priority
FR-CR01	All authenticated users can submit condition reports on assets they own. DEVELOPER is blocked from submitting on assets not assigned to them.	All	MUST
FR-CR02	DEVELOPER can only view their own submitted reports. ADMIN/MANAGER see all reports via AdminConditionReportView.	All	MUST
FR-CR03	ReportIssueForm Zod validation: issue (min 10 chars, non-blank), severity (enum: LOW MEDIUM HIGH).	All	MUST
FR-CR04	On 422 VALIDATION_ERROR, map error.details fields (issue, severity) to their respective form fields via setError.	All	MUST
FR-CR05	Condition report list must support URL-based filtering: status (OPEN IN_PROGRESS RESOLVED), severity (LOW MEDIUM HIGH), assetId, page, size.	ADMIN, MANAGER	MUST
FR-CR06	ResolveReportModal must validate allowed status transitions: OPEN -> IN_PROGRESS -> RESOLVED. Client must enforce these transitions.	ADMIN, MANAGER	MUST
FR-CR07	RESOLVED status update requires a non-empty resolution text field. Zod must enforce this.	ADMIN, MANAGER	MUST
FR-CR08	On successful submission (201), invalidate ["condition-reports"] and ["asset", assetId] query keys.	All	MUST
FR-CR09	On successful resolve/update, invalidate ["condition-reports"] and ["condition-report", id] query keys.	ADMIN, MANAGER	MUST

7. Reports & Analytics (reports)

Provides ADMIN and MANAGER roles with data-driven insights. DEVELOPER access is blocked — redirected to /dashboard.

7.1 API Endpoints

Method	Endpoint	Roles	Description
GET	/reports/dashboard	ADMIN, MANAGER	Aggregate KPI statistics: total assets by type/status, warranty alerts, open reports.
GET	/reports/usage	ADMIN, MANAGER	Asset usage & allocation history. Params: from, to (ISO date), type, userId.
GET	/reports/warranty-expiry	ADMIN, MANAGER	Assets with upcoming/past warranty expiry. Param: daysAhead (default 30).

7.2 Functional Requirements

ID	Requirement	Roles	Priority
FR-R01	reports/layout.tsx must block DEVELOPER role and redirect to /dashboard.	DEVELOPER	MUST
FR-R02	DashboardStatsPanel must display totalAssets, byType breakdown, byStatus breakdown, warrantyExpiringIn30Days, openConditionReports — all from GET /reports/dashboard.	ADMIN, MANAGER	MUST
FR-R03	UsageReportView must display: totalAllocations, averageAllocationDays, topUsers list, conditionReportsByMonth chart. Supports date range filter (from/to) via URL params.	ADMIN, MANAGER	MUST
FR-R04	WarrantyExpiryView must display asset list with daysUntilExpiry and suggestedAction (REASSIGN_AS_SPARE DECOMMISSION RENEW_WARRANTY). Each row links to the asset detail page.	ADMIN, MANAGER	MUST
FR-R05	daysAhead filter for warranty expiry must be configurable via URL param (integer). Default: 30.	ADMIN, MANAGER	SHOULD

8. Notifications (notifications)

In-app notification system for all authenticated users.

8.1 API Endpoints

Method	Endpoint	Roles	Description
GET	/notifications	All auth	Get user notifications. Filter by read status.
PATCH	/notifications/{id}/read	All auth	Mark a single notification as read.
GET	/notifications/preferences	All auth	Get notification preferences for current user.
PUT	/notifications/preferences	All auth	Update notification preferences.

8.2 Functional Requirements

ID	Requirement	Roles	Priority
FR-N01	NotificationBell component in Navbar must show an unread count badge (count of items where read === false).	All	MUST
FR-N02	NotificationBell must poll GET /notifications every 60 seconds via TanStack Query refetchInterval.	All	MUST
FR-N03	Clicking a notification item must call PATCH /notifications/{id}/read, optimistically or after response update the unread count.	All	MUST
FR-N04	Notification preferences must be editable: warrantyExpiryAlerts (bool), lowStockAlerts (bool), emailNotifications (bool), daysBeforeWarrantyAlert (int 1–365).	All	MUST
FR-N05	Zod validation for daysBeforeWarrantyAlert: integer in range 1–365.	All	MUST
FR-N06	Notification type WARRANTY_EXPIRY must render a link to the relevant asset detail page.	All	SHOULD
FR-N07	Unknown notification types must render a generic fallback (icon + message text only).	All	SHOULD

9. Role Permission Matrix

Feature / Action	DEVELOPER	MANAGER	ADMIN
Auth — Register / Login	YES	YES	YES
View own profile	YES	YES	YES
View all users list	NO	YES	YES
View other user profiles	NO	YES	YES
Manage user roles (PATCH)	NO	NO	YES
Delete users (DELETE)	NO	NO	YES
View own assets	YES	YES	YES
View all assets inventory	NO	YES	YES
Create / Update assets	NO	YES	YES
Delete / Decommission assets	NO	NO	YES
Allocate / Return assets	NO	YES	YES
View allocation history	NO	YES	YES
View spare laptops	YES	YES	YES
Asset free-text search	NO	YES	YES
Submit condition reports	OWN	YES	YES
Manage condition reports (PATCH)	NO	YES	YES
View reports & analytics	NO	YES	YES
Manage notifications & preferences	YES	YES	YES

10. API Error Code Reference

HTTP	Error Code	Meaning	Frontend Handling
400	BAD_REQUEST	Malformed body or missing fields	Show generic form error
401	UNAUTHORIZED	Missing or expired JWT	Clear auth, redirect to /auth/login
403	FORBIDDEN	Authenticated but wrong role	Redirect to /dashboard
404	*_NOT_FOUND	Resource does not exist	Show "not found" empty state UI
409	EMAIL_CONFLICT	Duplicate email on register	setError("email", ...)
409	SERIAL_NUMBER_CONFLICT	Duplicate asset serial number	setError("serialNumber", ...)
409	ASSET_ALREADY_ASSIGNED	Asset already deployed	setError("root", ...) in AllocationModal
409	NO_ACTIVE_ALLOCATION	Nothing to return	setError("root", ...) in AllocationModal
409	USER_HAS_ACTIVE_ALLOCATIONS	Cannot delete user with assets	Show specific message in DeleteUserModal
409	LAST_ADMIN_PROTECTION	Cannot remove last admin	setError("root", ...) in UpdateRoleModal
422	VALIDATION_ERROR	Semantic validation failure	Map error.details[] fields to setError(field, ...)
500	INTERNAL_ERROR	Unexpected server failure	setError("root.serverError", ...)